Did congestion pricing make NYC taxi trips faster? A dependable data pipeline from TLC records to a fleet decision

Ansh Mahajan · Roll 10345 · FDE Data Foundations, Assessment 2 (Track B - NYC TLC) · GitHub: https://github.com/AnshMahajan-grep/nyc-crz-taxi-pipeline

1. The problem worth solving (before any tooling)

On **5 Jan 2025** NYC began tolling vehicles entering Manhattan south of 60th St (the **Congestion Relief Zone, CRZ**). A yellow-taxi fleet operator must decide whether to **reposition drivers and re-plan shifts** around the zone. Leadership's claim is that *"the zone is faster now, so trips pay better"*. That claim is untested, and acting on it wrongly misallocates the ~70% of weekday trips that touch the zone. The raw data can't answer it directly. It is split across 4 technology vendors, 4-15% of records are reversals or meter errors, and the trip files have **no CRZ flag before 2025**. So the real problem is a **trustworthy, repeatable before/after measurement**, not a model or a dashboard.

Stakeholder	Decision / need
Fleet operations lead (client)	Reposition drivers toward CRZ trips in toll hours?
Finance / driver relations	Did revenue per engaged hour change?
TLC data owner	Which records we rejected and why (quality feedback)

Project KPI: median in-trip speed (mph) of trips that start *and* end inside the CRZ, on non-holiday weekdays, 05:00-20:59 (toll peak). Measured 5-31 Jan 2024 vs 5-31 Jan 2025, against a **control** group (Manhattan trips wholly north of the CRZ, not tolled).

2. Sources, retrieval and completeness

Source (owner)	Mode	Grain	How completeness is proven
Yellow Taxi trip records, monthly Parquet (NYC TLC; filed by 4 vendors)	File / HTTPS	trip record	bytes = Content-Length; rows loaded = Parquet footer (2,964,624 / 3,475,226); all 31 days present; >= 10k trips/day
Taxi zone lookup CSV (TLC)	File / HTTPS	zone (265)	>= 263 zones
Open-Meteo historical weather	REST / JSON	day	exactly one record per day
Nager.Date public holidays	REST / JSON	holiday	non-empty list
Modelled warehouse (this project)	SQL (SQLite)	trip/zone/day	rows after insert = validated rows; every metric is a SQL query

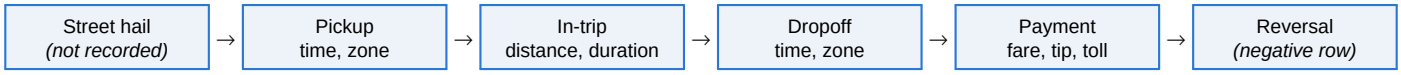
Raw inputs are stored unaltered with SHA-256 + row counts in *data/raw/manifest.json*. A rerun reuses a raw file only if its checksum still matches.

3. Profiling findings and validation rules (flag, never silently fix)

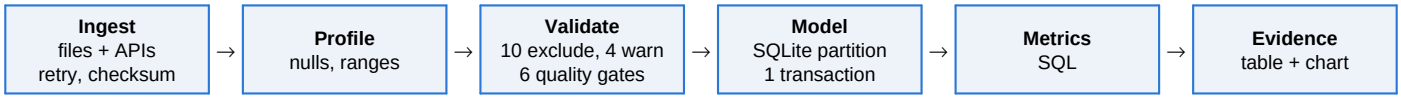
Finding (FACT, from profiling)	Jan 2024	Jan 2025	Rule / handling
Negative-fare rows: accounting reversals, not trips (83% have an exact mirror positive twin in 2024)	1.26%	4.15%	EXCLUDE; the twin is kept for speed but left out of revenue
payment_type 0 with null passenger/rate fields; durations and distances look normal	4.7%	15.5%	WARN, kept; sensitivity metric without them
Zero distance / < 60 s / > 3 h / > 80 mph / > 100 mi	2.5%	3.1%	EXCLUDE (meter/odometer/clock errors)
Unknown zone 264/265; out-of-month timestamps (e.g. year 2002)	1.1%	0.8%	EXCLUDE
Rows passing all rules	**95.5%**	**92.6%**	Quality gate: >= 85%, or month not published

Rows are never deleted. Every rule sets a bit flag in *fact_trip*, and counts per rule and example rows are published in *outputs/validation/*.

4. Workflow, data model and pipeline



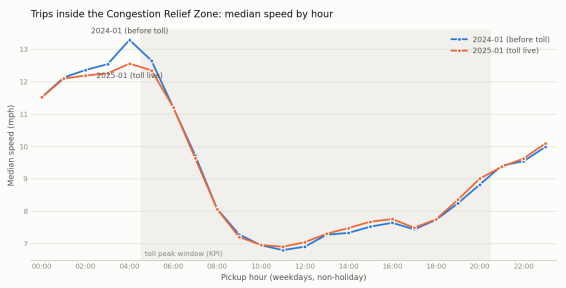
Intervention: CRZ toll (dim_date.is_toll_live, dim_zone.is_crz). **Confounders:** weather, holidays, weekends (dim_date). **Outcomes:** speed, p90 duration, fare/engaged hour, trip volume. **Star schema:** fact_trip (1 row per source record incl. invalid, dq/warn bitmasks, segment) - dim_zone - dim_date - metrics_monthly.



Dependability: stage logs + run_history.csv. A gate failure means that month is not published, and the evidence is regenerated only if *all* months pass. Exit code 1 on failure (tested with an unpublished month). Idempotent rerun: month partitions are replaced atomically. 17 unit tests.

5. Evidence: metrics linked to the KPI (5-31 Jan, weekday toll peak)

Metric	Jan 2024	Jan 2025	Change
Valid trip rate (trust)	95.51%	92.57%	-2.95 pp
KPI: CRZ median speed, weekday peak	7.74 mph	7.85 mph	+1.4%
Control: Manhattan outside CRZ	8.65 mph	8.45 mph	-2.3%
CRZ p90 trip duration	20.8 min	20.1 min	-3.6%
CRZ fare per engaged hour	$63.07	$63.86	+1.3%
CRZ-touching trips / weekday	68,057	77,918	+14.5%
KPI, dry days only	7.83 mph	7.83 mph	+0.1%
Control, dry days only	8.75 mph	8.46 mph	-3.3%
KPI excl. payment_type 0	7.72 mph	7.80 mph	+1.0%



Net KPI effect vs control: +3.7% (all days), +3.4% (dry days only). Raw CRZ speed rose only 1.4%, but the untolled control slowed 2.3%. January 2024 had 12 wet days (156 mm) against 5 (23 mm) in 2025. On dry days the CRZ is flat while the control slowed 3.3%, so the *relative* effect survives the weather check and the raw gain does not. The result also holds without payment_type 0 records.

6. Conclusion and the decision it supports

Question	Answer (evidence)
Are zone trips faster?	Barely in absolute terms (+1.4%, < 1 min per trip). The toll's effect is an **avoided slowdown** of about 3-4% relative to the rest of Manhattan
Did the toll reduce taxi demand in the zone?	No: +14.5% trips/weekday, in line with overall valid-trip growth (+13.6%)
Did drivers earn more per hour?	Essentially flat: $63.07 to $63.86 per engaged hour (+1.3%)
Decision	**Keep or grow driver supply in the CRZ during toll hours** (demand and revenue held). **Don't** plan shifts on the assumption that zone trips are faster. Rerun monthly (*python -m pipeline.run --months ...*) before changing the plan.

7. Facts, assumptions, unknowns and limitations

Type	Statement
FACT	Both trip files are complete (bytes, footer row counts, 31/31 days); all 6 quality gates passed
FACT	Our 38-zone CRZ list agrees with the 2025 toll flag on 93.2% of intra-zone trips; every listed zone is tolled on >= 94%, and no unlisted zone above 42%
FACT	Jan 2025 was far drier than Jan 2024 (23 mm vs 156 mm), so weather is a real confounder
ASSUMPTION	The CRZ is approximated by whole taxi zones. Zones straddling 60th St (Lincoln Sq, Lenox Hill, Central Park) are counted as outside
ASSUMPTION	Upper Manhattan is a fair control. If it also gained from less through-traffic, our net effect is a lower bound
ASSUMPTION	Jan 2024 is a valid counterfactual after matching weekday, holiday, peak hour and (sensitivity) weather
UNKNOWN	Customer wait/dispatch time (street hails aren't recorded); whether trips passed through the CRZ (no route)
UNKNOWN	Why payment_type 0 records tripled - to be raised with TLC/vendors
LIMITATION	One month before and after; yellow taxis only (not Uber/Lyft); no significance test; revenue = metered fare per engaged hour (tips only recorded for card payments; no driver IDs, so no idle time)

8. The FDE judgement call

The obvious shortcut is the new *cbd_congestion_fee* column. But it only exists **after** the intervention, so it can't define a 2024 baseline, and any before/after comparison built on it would be invalid. I defined the CRZ from zone geography instead, as an explicit, challengeable config list applied identically to both years. I then turned the source's own toll flag into a **quality gate** that re-tests the definition on every 2025+ run. If the list is wrong or TLC re-draws zones, the pipeline refuses to publish. The assumption is visible, tested and enforced, rather than hidden.

Repository: https://github.com/AnshMahajan-grep/nyc-crz-taxi-pipeline · **Demo:** Loom link submitted in the form · Reproduce: *pip install -r requirements.txt; python -m pipeline.run*